

GM1 report - Hassan Himaz:

Week 1

This report details week 1 activity. I primarily focussed on mechanical and electrical design, with some work on system architecture and decision making (like gait classification and force sensing) to better understand mechanical requirements. For brevity and relevance, I have focussed on activities relating to design decisions, as opposed to descriptions of attempts at fabrication and bug fixing.

TIME BREAKDOWN:

1. Design of mechanical force sensing [15 hours]
2. Setup of electronics and Sseeed Board [5 hours]
3. Software considerations to inform mechanical design. [5 hours]

MECHANICAL AND ELECTRICAL DESIGN:

This week we tried to get force sensing working.

Magnetic sensing:

We had opted for a lower cost hall effect setup. This was at the expense of higher mechanical complexity and longer setup times but, compared with buying an off the shelf load cell, should result in lower costs and a sleeker form factor.

We had planned to use a spring with 5mm of deflection, as we were unsure of the noise of the magnetic sensor, and background noise. But upon testing we realised we had less noise than anticipated so could use lower deflection. We aimed to be able to differentiate between 10kg steps at about 50kg.

Initially we wanted to work with very rough estimates of values, for instance we observed the noise was ± 10 (digital output of magnetometer) and then calculated the number of steps we would need in the digital output magnetic field to meet the required resolution. As we struggled with finicky practical engineering issues (detailed later). It became apparent that a more systematic approach was needed. This approach is detailed in appendix 1. Using this method, and from experimental results, we now aim to get about 0.5mm of compression distance. As we have a variety of magnets of different strengths (and unknown composition) the exact distance between the magnet and sensor is unknown. But given tests a distance of 15-25 mm seems reasonable.

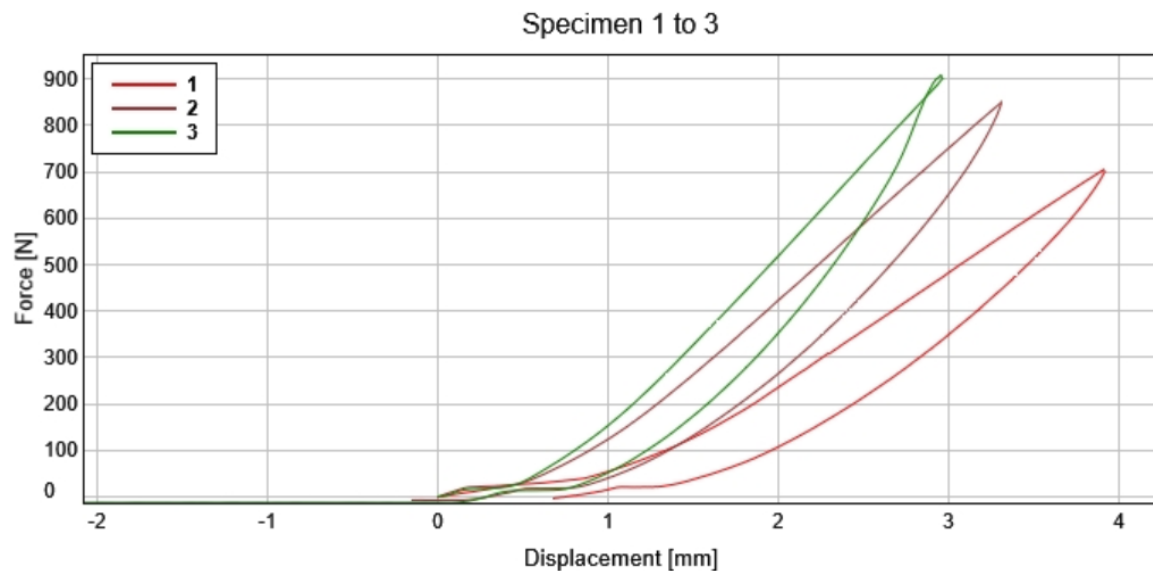
Compression Element:

As we now needed less compression we could switch to an elastomer. This had the advantage of simpler construction (no guide rails needed), better durability (no water/mud proofing), and better resistance to shear forces/bending moments. This has the downside of non-linearity and a variable strain rate. Essentially force would be a function of displacement and velocity. This is based on Visco-elastic theory, essentially a nonlinear in parallel with a damper.

We had rubber bungs and sheets, but to improve simplicity of design and fabrication decided we would 3D print a foot out of TPU. In the production version this would be injection moulded/ cast for greater consistency. Though we got better than expected sensitivity to force from the magnet, the variable strain rate was greater than we expected, meaning we would have had to implement

a 2d lookup table. If we were to include temperature, we could either extend the table or assume a scaling factor.

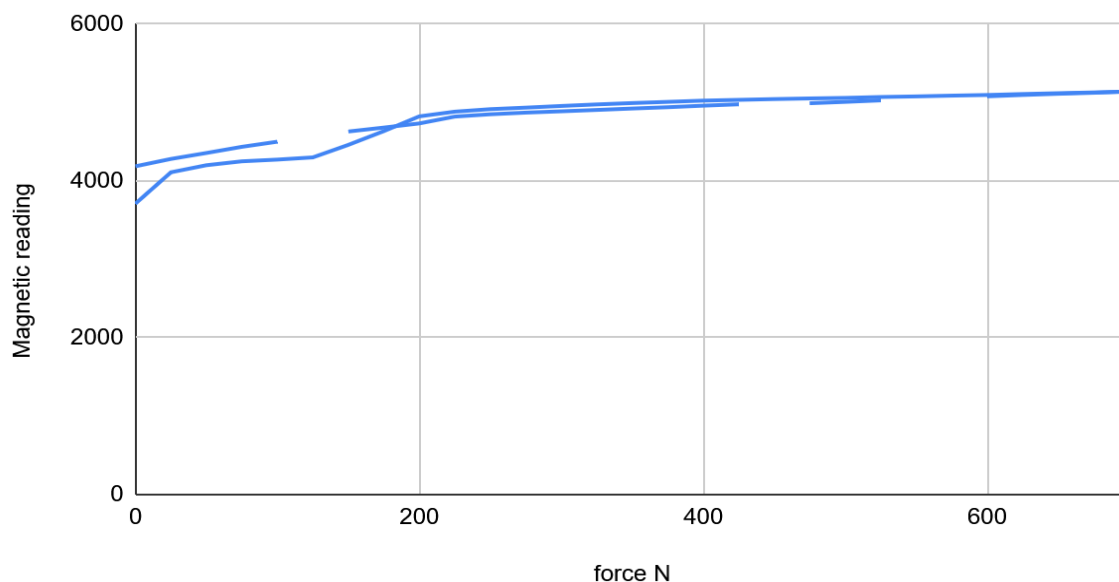
We can see this issue in greater detail from this graph of stress-strain, obtained on the Instron machine for: 1: 0.5mm/s, 10mm/s, 100mm/s. The speed of impact changes over its duration but rough calculations indicate it will be in the 100mm/s order of magnitude (based on measured impact times)



Notice the initial nonlinear region - this is due to the foot not being totally flat with the plate: an effect of deformations in 3d printing.

We also present a graph of magnetic reading against force measure; this is strangely linear. The top line without gaps is loading, and the line below that with some gaps (reading was changing too fast to note down) is unloading.

Magnetic reading vs force N



The results with the TPU rig were disappointingly nonlinear, likely due to the complexity of the 3d mesh. This led to us falling behind on our schedule - In retrospect we should have been more generous as to things not going right the first time.

SOFTWARE CONSIDERATIONS:

Concurrently with mechanical design, we worked closely with the software team, both in helping everyone understand the integrated system, and in evaluating force sensing to better constrain the mechanical design.

I spent a couple of hours segmenting the types of crutch gaits: one legged, two-point, 3-point, 4 points; to understand how to detect and evaluate them. We concluded that detection could be achieved just with the IMUs and long-term force sensing (to assess recovery and improper use on injured feet) could be found just with peak sustained load on the crutch. Sensor fusion was not required, and it was agreed that a 10 kg range of accuracy was fine.

Later, we also decided it would be useful to have a result of how the force changed during impact, though absolute accuracy was not required. A high sampling rate (not yet specified - likely about 100Hz to fuse with IMU data) would be useful to accurately analyse the shape of the impact.

DECIDED FUTURE:

After the setbacks on Thursday, we held a team meeting to recalibrate. Key outcomes from this:

1. 3D printed TPU may increase the variable rate issue - so pivoting to premade rubber/silicon sheet could mitigate the issue.
2. If sheet rubber still has these issues, then the best solution would be to test over the range of values and make a 3d lookup table of displacement, velocity, and temperature.
3. A spring, or cantilever, would solve our non-linearity problems.

Actions:

1. Build new modular design to test more rubbers and potential cantilevers. Modular design should be able to easily switch from testing mode (flat plate on the top) to being attached to the crutch. Modular design would consist of 3 layers - a top layer to hold the magnetometer, a compression layer, and a foot layer to hold the magnet. We should be able to adjust magnet type and distance via a stackable design.
2. Test built rubbers in Instron on Wednesday.

NEXT WEEK AIM:

Build a working prototype by Thursday. Achieve this through a modular design, with Instron testing and generation of lookup tables to happen on Wednesday (Earliest machine is free next week).

RISKS:

Non linearities in the sensor make force sensing unviable (low chance, large impact)

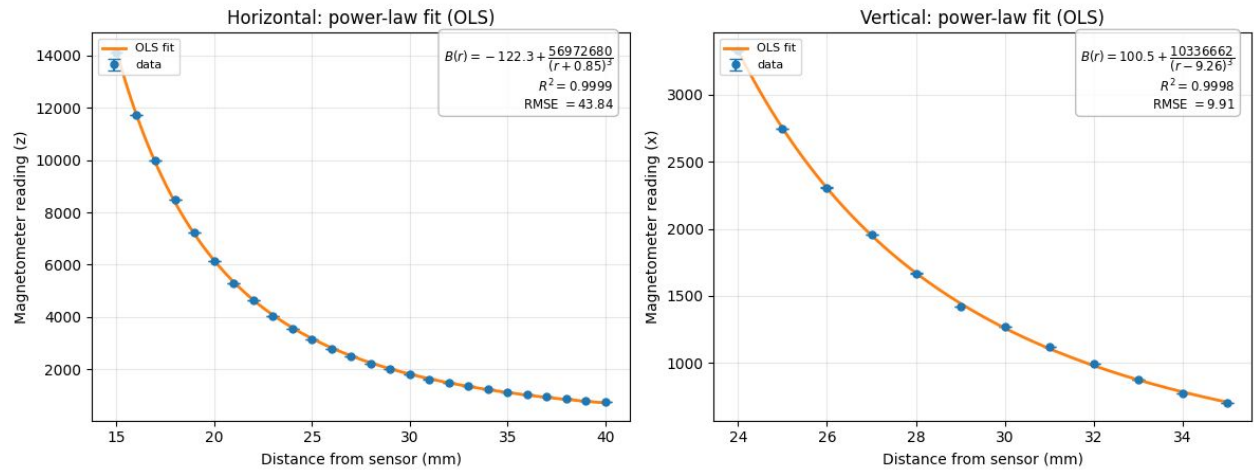
Lack of force sensing early means software has compressed development time.

To mitigate the risks, we plan to look into developing some compression elements in parallel - so we pursue both a cantilever and rubber.

APPENDIX 1 - MAGNETIC FIELD CONSIDERATIONS:

Testing and theoretical analysis suggested that the magnetic field detected by the sensor followed an inverse cube law:

$$\text{OLS} - B(r) = B_0 + A / (r + r_0)^3$$



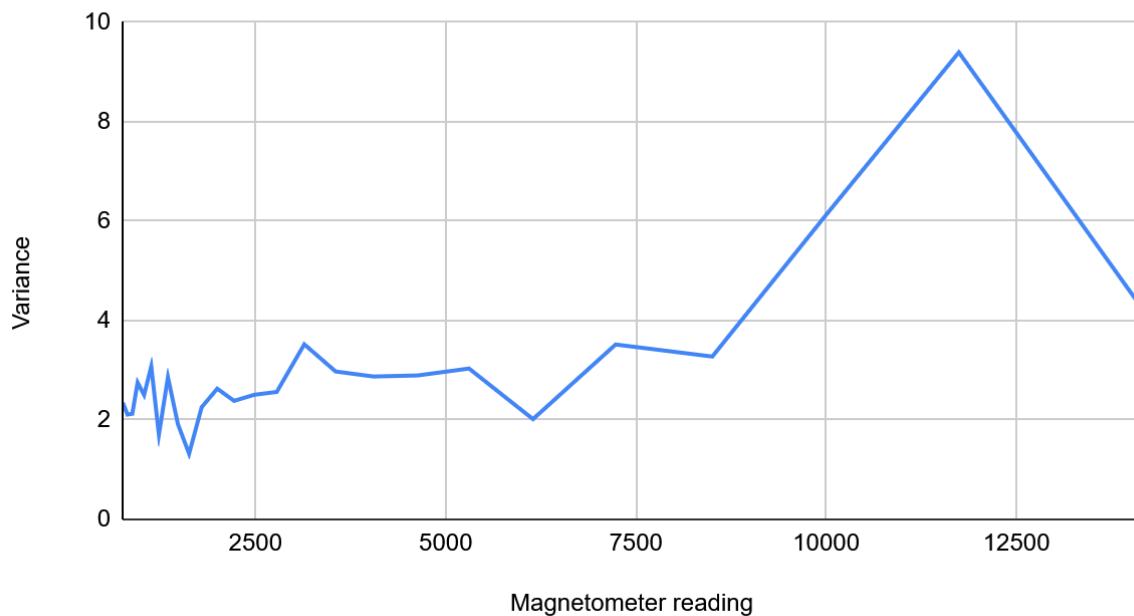
Then we can calculate the sensitivity of the reading to noise by differentiating the law:

$$\left| \frac{dB}{dr} \right| = \frac{3B}{r}$$

Note that to get this result we substitute the B for the constant in the differentiation. If we did not do this, we would get an inverse quartic in R. Basically saying that the closer we are to the magnet the greater the sensitivity, which is obvious.

If we then consider the variance of the sensor with regard to the magnetometer reading:

magnetometer reading (z) and Variance from 50 samples



We can see that up to 8000 the variance is about 3. Then to find how small a distance we can discern we do noise/Sensitivity:

$$\text{Discernable step} = \frac{\text{Noise}}{\text{Sensitivity}} = \frac{\text{Noise} \times R}{3B} = \frac{\text{Noise} \times R^4}{k}$$

We can see from this equation that we should minimise the distance from the magnet. Our constraint is the range of the magnetic sensor so theoretically we should pick a magnet that maxes out the range when the sensor is as close as possible to the magnet (now limited by compression material). An issue with this is that we don't know how noise changes with all magnet types so this calculation may not be correct. Given our limited timescale, and the fact that we know 0.5mm of displacement paired with a sensor that is 15mm away works well enough, it would make sense to hold off optimising this distance until we rid ourselves of the more significant compression issues.